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Method and system for pressure relief in a multi chamber vessel

Abstract

A vessel includes a body that defines an interior space. The body includes a first metal surface, a second metal surface, and a non-metallic material crimped between the first metal surface and the second metal surface at a joint. The vessel further includes a pressure relief device coupled to the body. The pressure relief device is configured to vent contents of the vessel to an exterior of the body. Upon exposure of the vessel to a temperature for a period of time, the second metal surface is configured to expand or bend to create a pressure relief route from the interior space to the pressure relief device, between the first metal surface and the second metal surface, and/or the non-metallic material is configured to melt to create the pressure relief route.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims the benefit of U.S. Non-provisional application Ser. No. 17/727,251 filed Apr. 22, 2022 and entitled “METHOD AND SYSTEM FOR PRESSURE RELIEF IN A MULTI CHAMBER VESSEL”. The entirety of the aforementioned application is incorporated herein by reference.

BACKGROUND

(1) Vessels can be used for storage and/or dispensing of gasses, liquefied compressed gasses or liquids. Certain vessels can include multiple internal chambers separated by a flexible membrane, for example, a diaphragm, a bladder, or a bag. In vessels where one of the chambers does not have a means for relieving excess pressure, dangerous over-pressure situations can occur. For example, high pressure situations may occur due to a fire, excess heat, or overfilling. Such high-pressure situations may cause dangerous consequences such as explosions and catastrophic ruptures creating projectiles, which may injure people and damage equipment.

SUMMARY

(2) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key factors or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

(3) One or more techniques and systems described herein can be utilized to provide a vessel that includes a body having an interior surface that defines an interior space. A flexible membrane such as a diaphragm, a bag, or a bladder can be located within the interior space of the vessel. The interior space can include a first chamber at least partially defined by the flexible membrane, and a second chamber at least partially defined by the flexible membrane and a portion of the interior surface of the body. A pressure relief device can also be coupled to the body. The pressure relief device is configured to vent contents of the second chamber to an exterior of the body when the second chamber reaches a first predefined pressure. The flexible membrane is configured to tear or puncture when the first chamber reaches a second predefined pressure that is less than or equal to the first predefined pressure. In this manner, dangerously high pressures in the first chamber can be prevented.

(4) In one embodiment, the vessel includes a piercing member extending into the second chamber. The piercing member is designed and positioned to puncture the flexible membrane when the first chamber reaches the second predefined pressure.

(5) In another embodiment, the flexible membrane is constructed and sized such that the flexible membrane rips or tears upon the first chamber reaching the second predefined pressure.

(6) In another aspect, the vessel can be constructed with walls that include one or more non-metallic materials sandwiched between two metallic layers. In a situation where excess heat or flame is applied to the vessel, thermal expansion of the metallic layers and/or contraction or softening of the one or more non-metallic layers create a pressure relief channel. The pressure relief channel allows for contents from the interior of the vessel to vent from the interior through the pressure relief channel, and through the pressure relief device to the exterior of the vessel.

(7) To the accomplishment of the foregoing and related ends, the following description and drawings set forth certain illustrative aspects and implementations. These are indicative of but a few of the various ways in which one or more aspects may be employed. Other aspects, advantages and novel features of the disclosure will become apparent from the following detailed description when considered in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a component diagram illustrating an example embodiment of a vessel, where one or more techniques and/or one or more systems described herein may be implemented;
- (2) FIG. 2 is a top perspective view of an example embodiment of a vessel;
- (3) FIG. 3 is a bottom perspective view of an example embodiment of a vessel;
- (4) FIG. 4A is an enlarged view of an exemplary piercing member;
- (5) FIG. 4B is a cross sectional view of an exemplary pocket;
- (6) FIG. 5A is a component diagram illustrating another example embodiment of a vessel;
- (7) FIG. 5B is a component diagram illustrating another example embodiment of a vessel;
- (8) FIG. 6 is a flow chart depicting an exemplary method of constructing a vessel; and
- (9) FIG. 7 is a cross-sectional view of an example embodiment of a vessel wall.

DETAILED DESCRIPTION

(10) The claimed subject matter is now described with reference to the drawings, wherein like reference numerals are generally used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the claimed subject matter. It may be evident, however, that the claimed subject matter may be practiced without these specific details. In other instances, structures and devices are shown in block diagram form in order to facilitate describing the claimed subject matter.

(11) A vessel, as used herein, is defined as any container capable of storing and/or dispensing contents contained within. By way of example and not limitation, a vessel, as used herein, may be a pressure vessel.

(12) A vessel can include two or more chambers separated by a flexible membrane such as a diaphragm, a bag, or a bladder. The vessel can include a pressure relief device located on the body of the vessel. This pressure relief device can function to relieve/vent pressure from one chamber that is in direct fluid communication with the pressure relief device. However, the pressure relief device is unable to relieve/vent pressure from the other one or more chambers while the flexible membrane is intact. Rather, the vessel can include one or more means to tear or puncture the flexible membrane so that the pressure of the one or more other chambers may also be relieved and vented to the exterior of the vessel through the pressure relief device.

(13) Turning now to FIG. 1, a vessel **100** is shown. The vessel **100** can be, for example, a storage tank or an expansion tank such as a diaphragm tank or a bladder tank, among others. Vessel **100** includes a body **102** that can be constructed of any material chosen using sound engineering judgment. By way of example, and not limitation, the body **102** can be constructed using one or more of metal, such as steel or aluminum, carbon fiber, glass fiber, or a polymer such as high density polyethylene. The body **102** includes an interior surface **104** that defines an interior space of the vessel **100**.

(14) The vessel **100** can further include a flexible membrane **106** located within the interior space of the vessel **100**. The flexible membrane **106** can form distinct chambers within the interior space of the vessel **100**. In one embodiment, a first chamber **108** can be at least partially defined by the flexible membrane **106**. A second chamber **110** can be at least partially defined by the flexible membrane **106** and a portion of the interior surface **104** of the body **102**. It should be appreciated that certain embodiments of the vessel **100** can include additional flexible membranes and/or additional chambers. During normal operation of the vessel **100**, the flexible membrane **106** causes the first chamber **108** and the second chamber **110** to be fluidly isolated from each other such that there is no fluid communication between the first chamber **108** and the second chamber **110** within the interior space of the vessel **100**. By way of example, and not limitation, the flexible membrane **106** can be constructed from an elastomer (e.g. ethylene propylene diene monomer (EPDM), butyl, nitrile, neoprene or silicone rubber), a film such as a polyester film (e.g. biaxially-oriented

polyethylene terephthalate (BoPET)), which may be a single-ply film or a multi-ply film, or a foil.

(15) The vessel **100** can further include a pressure relief device **112** coupled to the body **102**. The pressure relief device **112** can be, for example, a pressure relief valve, a burst disc, or any other pressure-activated relief mechanism. The pressure relief device **112** can be located on the vessel **100** body **102** on the perimeter of the second chamber **110**, and can be configured to vent the contents of the second chamber **110** to the exterior of the body **102** when the second chamber **110** reaches a first predefined pressure. The first predefined pressure can be selected to prevent excess pressure from building within the vessel **100** and damaging the vessel **100** or causing a dangerous explosion or projectile. In one embodiment, the first predefined pressure can be selected to be greater than or equal to 325 pounds per square inch (psi). In another embodiment, the first predefined pressure can be selected to be greater than or equal to 112 psi. In one embodiment, when the pressure within the second chamber **110** reaches the first predefined pressure, the pressure relief device **112** opens to vent the contents of the second chamber **110** to the exterior of the vessel **100** and cause the second chamber **110** to have ambient pressure. It should be appreciated that the first predefined pressure can be a particular pressure, a particular pressure with a tolerance, or a range of pressures.

(16) In certain embodiments, the vessel can include a piercing member **114**. The piercing member **114** extends into the second chamber **110** towards the flexible membrane **106**. In different embodiments, the piercing member **114** can extend directly from an interior surface **104** of the body **102** into the second chamber **110**, extend directly from the pressure relief device **112**, extend from a standoff **115** that separates the piercing member **114** from the pressure relief device **112** by a distance, or extend from another component of the vessel **100**. The piercing member **114** can be, for example, one or more razors, burrs, serrated surfaces, needles, or other sharp or rough surface.

(17) The piercing member **114** is configured to puncture the flexible membrane **106** when the first chamber **108** reaches a second predefined pressure. In certain embodiments, the second predefined pressure is less than or equal to the first predefined pressure at which the pressure relief device **112** operates to vent the second chamber **110**. During normal operation when the pressure relief device **112** is not activated, the first chamber **108** and the second chamber **110** share a common pressure. If the flexible membrane **106** remains intact long after the pressure relief device **112** opens and begins venting the second chamber **110**, there is a risk that the flexible membrane **106** can continue to expand and seal or significantly restrict the pressure relief device **112**, thereby preventing the vessel **100** from being properly vented. In this situation, the dangerous high pressure can build again within the vessel **100**. Accordingly, the flexible membrane **106** and/or piercing member **114** can be designed to cause the flexible membrane **106** to tear or puncture prior to making contact with the pressure relief device **112**. To achieve this end, the second predefined pressure can be chosen to be less than or equal to the first predefined pressure. It should be appreciated that the second predefined pressure can be a particular pressure, a particular pressure with a tolerance, or a range of pressures.

(18) Other features may be included within the vessel **100** to prevent the flexible membrane **106** from sealing the pressure relief device **112**. For example, a portion of the interior surface **104** within the second chamber **110**, a surface of the flexible membrane **106**, or both can include one or more projections. These projections can include one or more of channels, bumps, ridges, or knurls, among others. The projections can prevent a complete seal from being formed in the event that the flexible membrane **106**, or a portion thereof, happens to make contact with the pressure relief device **112** or any aperture that provides access to the pressure relief device **112**.

(19) In certain embodiments, the vessel **100** can include a pocket **116** formed in the body **102** of the vessel **100**. The pocket **116** can be formed in the body **102** such that the pocket extends inwards into the second chamber **110**. For example, the pocket **116** can include a convex shape in the body **102**, indented within the second chamber **110**. The standoff **115** can extend from the pocket **116** into the second chamber **110**, and the piercing member **114** can extend from the standoff **115** into

the second chamber **110**, as shown in FIG. 1. The pressure relief device **112** can be installed into the pocket **116**. For example, the pressure relief device **112** can fit into and be mounted on an exterior of the indent formed by the pocket **116**.

(20) As an example of operation of an embodiment of the vessel **100**, the pressure increases within the vessel **100**. The vessel's **100** pressure increase causes the pressure to increase in both the first chamber **108** and the second chamber **110**. As the pressure increases in the first chamber **108**, the flexible membrane **106** expands into space previously occupied by the second chamber **110**. It should be appreciated that the body **102** maintains its shape as the flexible membrane **106** expands. The flexible membrane **106** continues to expand as the pressure of the first chamber **108** increases towards the second predefined pressure, and the flexible membrane **106** approaches the piercing member **114**. As the pressure of the first chamber **108** reaches the second predefined pressure, the flexible membrane **106** contacts the piercing member **114** with enough force to be punctured by the piercing member **114**. Simultaneously, or soon after the flexible member **106** is punctured by the piercing member **114**, the second chamber **110** reaches the first predefined pressure, which causes the pressure relief device **112** to open.

(21) In one embodiment, the flexible membrane **106** is configured to be punctured and/or tear at a second predefined pressure, but only after the pressure relief device **112** opens at the first predefined pressure. In this situation, both the first chamber **108** and the second chamber **110** can reach the first predefined pressure. The pressure relief device **112** is configured to open at the first predefined pressure, which causes the pressure of the second chamber **110** to drop to 0 psi. Simultaneously with the decrease in pressure of the second chamber **110**, the flexible membrane **106** expands, causing the pressure of the first chamber **108** to also decrease to less than the first predefined pressure at which the pressure relief device **112** opened. Over time, as pressure continues to build in the first chamber **108**, the flexible membrane **106** continues to expand and contacts the piercing member **114** with enough force to puncture the flexible membrane **106**.

(22) In one embodiment, the flexible membrane **106** is configured to tear when the first chamber reaches the second predefined pressure. The strength, thickness, and type of material can be selected to ensure that the flexible membrane **106** fails when the first chamber **108** reaches the second predefined pressure. It should be appreciated that while this embodiment may not require a piercing member **114** to puncture the flexible membrane **106**, a piercing member **114** may still be included as a backup to ensure that the flexible membrane **106** is punctured in the event that the flexible membrane **106** does not tear as expected.

(23) The vessel **100** can further include components such as a valve **118** coupled to the vessel **100**, and in fluid communication with the first chamber **108**. A shroud **120** can also encircle the valve **118** to provide protection to the valve **118** and also provide handles for grasping and lifting the vessel **100**. A propellant valve **122** can also be coupled to the body **102** to provide a selective fluid communication between the interior space of the vessel **100** and a source of a propellant used to create a pressure differential between the second chamber **110** and the exterior of the vessel **100** so that the contents of the first chamber **108** can be expelled out of the valve **118**. In one embodiment, the propellant valve **122** provides selective fluid communication between the second chamber **110** and a source of a propellant. In this embodiment, the propellant valve **122** can be utilized by a user for filling the second chamber **110** with a pressurized gas such as pressurized air, nitrogen, carbon dioxide, or compressed liquified gas such as propane, butane or refrigerant, among others. The propellant valve **122** can extend into the second chamber **110**. In certain embodiments, the propellant valve **122** can include a piercing member **114** extending therefrom into the second chamber **110** towards the flexible membrane **106**. In these embodiments, the propellant valve **122** can include the sole piercing member **114**, or it could also include an auxiliary piercing member **114** in addition to a primary piercing member **114** located elsewhere on the vessel **100** (e.g. on the pressure relief device **112**, pocket **116**, or standoff **115**). In another embodiment, a portion of the propellant valve **122** that extends into the second chamber **110** is a piercing member **114**.

(24) With further reference to FIG. 2 and FIG. 3, the body **102** can be constructed from a first portion **124** and a second portion **126** that are connected by, for example, welding, brazing, crimping, or flange. The connection of the first portion **124** and the second portion **126** can create a seam **128**. In certain embodiments, the flexible membrane **106** can be crimped to at least one of the first portion **124** or the second portion **126** of the vessel **100** body **102**. In certain embodiments, the flexible membrane **106** is crimped between the first portion **124** or the second portion **126**, and a crimping ring **130**, which extends around the inner circumference of the vessel **100** body **102**. In other embodiments, the body **102** of vessel **100** can be seamless.

(25) In certain embodiments, both of the first portion **124** and the second portion **126** include a convex shell and a cylindrical sidewall portion. The first portion **124** can be configured to receive the valve **118**, which provides selective fluid communication between the first chamber **108** and the exterior of the body **102**. The valve **118** can be used either for filling the vessel **100**, dispensing contents of the vessel **100**, or both. A shroud **120** can also be mounted to the first portion **124**. The second portion **126** can include one or more of the pressure relief device **112**, the piercing member **114**, and the propellant valve **122**. The second portion **126** can further include a stand **132** such as a footing or plurality of dimples for feet on which the vessel **100** can stand. The stand **132** provides balance and support for the vessel **100**. In one embodiment, the pressure relief device **112** and the piercing member **114** are both located on the convex shell of the second portion **126**. Specifically, any weldments and openings for the pressure relief device **112** and/or the piercing member **114** are both located within the innermost 80% portion of the radius of the convex shell, if located on the convex top portion of the second portion **126**. There are no location limitations, however, if pressure relief device **112** and/or piercing member **114** are on the side walls of the second portion **126**. It should be appreciated that the vessel **100** can be operable while positioned in any orientation, including with the first portion **124** oriented upwards and the second portion **126** oriented downwards, the first portion **124** oriented downwards and the second portion **126** oriented upwards, or the vessel **100** oriented on its side such that the first portion **124** and the second portion **126** extend along a common horizontal plane.

(26) Turning now to FIGS. 4A and 4B, an exemplary piercing member **114** is shown. The piercing member **114** can extend from the standoff **115**, which separates the piercing member **114** from the pressure relief device **112** by a vertical distance **134** perpendicular to the interior surface **104** of the vessel **100**. By separating the piercing member **114** from the pressure relief device **112** by the distance **134**, the likelihood of the flexible membrane **106** contacting or sealing the pressure relief device **112** while the pressure relief device **112** is venting the second chamber **110** to the exterior of the vessel **100** is reduced. The standoff **115** can extend from the pocket **116**, which at least partially surrounds the pressure relief device **112**. In one embodiment, the pocket **116** includes a convex surface extending into the second chamber **110**, which also reduces the likelihood of the flexible membrane **106** contacting or sealing the pressure relief device **112** while the pressure relief device **112** is venting the second chamber **110** to the exterior of the vessel **100**. As shown in the cross-sectional view of FIG. 4B, the pressure relief device **112** can be mounted at least partially within the indent formed by the pocket **116**, on an exterior of the body **102**.

(27) The piercing member **114** and the standoff **115** can form an aperture **136** that defines a passageway that provides fluid communication between the pressure relief device **112** and the second chamber **110**. In one embodiment, the aperture **136** is at least 10 millimeters in diameter to reduce the likelihood of the flexible membrane **106** from restricting or sealing the pressure relief device **112** when venting the first chamber **108** and/or second chamber **110** to the exterior of the vessel **100**. The piercing member **114** can include a plurality of piercing points **138** surrounding the aperture **136**. Each of the plurality of piercing points **138** can be separated from each other by a space. In certain embodiments, the piercing member **114** can be formed from the standoff **115**. In these embodiments, the piercing member **114**, including any piercing points **138**, project directly from the standoff **115**.

(28) FIG. 5A depicts another exemplary embodiment of a vessel **500**. It should be appreciated that the construction of vessel **500** can be similar to that of vessel **100** and include two portions connected together at a seam, or alternatively be seamless. The vessel **500** includes a flexible membrane **506**. The flexible membrane **506** is in the form of a bladder or bag that completely defines the first chamber **508**. The flexible membrane **506** can be attached to the valve **518** within the vessel **100** or at the junction of the valve **518** and the body **502**. The second chamber **510** exists as the area outside of the flexible membrane **506**, defined by the inner surface **504** of the vessel **500** and the outside of the flexible membrane **506**. Similarly as the embodiments described above with respect to vessel **100**, vessel **500** can include a pressure relief device **512** in the body **502** of the vessel **500**, and a piercing member **514** that can extend from a standoff **515** that spaces the piercing member **514** from the pressure relief device **512**.

(29) Similar to vessel **100**, vessel **500** can include a pocket **516** formed in the body **502**. The standoff **515** can extend from the pocket **516** into the second chamber **510**, and the piercing member **514** can extend from the standoff **515**. Vessel **500** can further include a valve **518** and a propellant valve **522** that serve the same functions as in vessel **100** as described with respect to FIG. 1. The valve **518** can be at least partially surrounded by a shroud **520**. It should be appreciated that the propellant valve **522** may also be at least partially surrounded by a second shroud.

Propellant valve **522** can be located on an end of the vessel **500** (as shown) or it can alternatively be located on a side wall of the vessel **500**. If the propellant valve **522** is mounted to either end of the vessel **500**, it may be mounted within the innermost 80% portion of the radius of the convex shell that makes up an end of the vessel **500**. Vessel **500** can further include a stand (not shown in FIG. 5A) such as a footring or a plurality of dimples for feet on which the vessel **500** can stand. The stand can be included on either end of the vessel **500**, and any valve or shroud (e.g. valve **518**, shroud **520**, propellant valve **522**) may be sized, positioned, or omitted accordingly to allow the stand to function without obstruction. For example, a stand can be included near the propellant valve **522**. In this case, the propellant valve **522** can be sized to not extend past the stand, and a shroud around the propellant valve **522** can be omitted. In another example, a shroud (e.g. shroud **520**, or a shroud around the propellant valve) can also function as a stand.

(30) FIG. 5B depicts another exemplary embodiment of a vessel **550**. Vessel **550** includes a first flexible membrane **556a** that at least partially defines a first chamber **558a**, and a second flexible membrane **556b** that at least partially defines a second chamber **558b**. The vessel **550** also includes a third chamber **560** at least partially defined by the body **552** of the vessel **550**. Vessel **550** can include a first valve **568a** that provides selective fluid communication into and out of the first chamber **558a** and a second valve **568b** that provides selective fluid communication into and out of the second chamber **558b**. In certain embodiments, the first valve **568a** and the second valve **568b** are integrated into a single valve body. A shroud **570** can surround both the first valve **568a** and the second valve **568b**, or the vessel **550** can include a shroud **570** for each of the first and second valves **568a**, **568b**. Vessel **550** can include at least one of a first piercing member **564a**, a first standoff **565a**, and/or a first pocket **566a**, which correspond to the first flexible membrane **556a**, and function in the same manner as described in the other embodiments above. For example, the first piercing member **564a** can be arranged and configured to puncture the first flexible membrane **556a** when the first chamber **558a** reaches a particular pressure. Further, vessel **550** can include at least one of a second piercing member **564b**, a second standoff **565b**, and/or a second pocket **566b**, which correspond to the second flexible membrane **556b**, and function in the same manner as described in the other embodiments above. For example, the second piercing member **564b** can be arranged and configured to puncture the second flexible membrane **556b** at a particular pressure of the second chamber **558b**. The vessel **550** can include at least one pressure relief device **562a**, **562b** to selectively vent pressure from the third chamber **560** to the exterior of the vessel **550**. For example, a first pressure relief device **562a** can be mounted within the first pocket **566a** and/or the second pressure relief device **562b** can be mounted within the second pocket **566b**.

(31) Turning now to FIG. 6, a method for constructing a vessel **100** is depicted. At reference numeral **602**, a body **102** is provided. At reference numeral **604**, one or more flexible membranes **114** are coupled within an interior of the body **102** to create at least a first chamber **108** and a second chamber **110** within the body **102**. For example, the flexible membrane **106** can be crimped to at least one of the first portion **124** or the second portion **126** with a crimping ring **130**, or the flexible membrane **106** can attach to a valve **118** coupled with the body **102**. At reference numeral **606**, a pocket **116** is formed in the wall of the body **102**. In one embodiment, the pocket **116** can be formed to include a convex shape indented within the second chamber **110**. In certain embodiments, the pocket **116** can be formed using a die. At reference numeral **608**, a hole is punched in the pocket **116** to form a standoff **115** and a piercing member **114** that includes an aperture **136** and piercing points **138** surrounding the aperture **136** and projecting from the standoff **115**. The aperture **136** can act as an entrance to a passageway that provides fluid communication between a pressure relief device **112** and the second chamber **110**. For example, in an embodiment where the pocket **116** has a convex shape, the hole is punched from the inside of the convex shape (corresponding to the exterior of the vessel **100**) outwards so that the piercing points **138** extend outwards from the convex surface of the pocket **116**. In other words, the punching force is directed inwards towards the interior of the body **102**. In one embodiment, the aperture **136** is created to be at least 10 millimeters in diameter. At reference numeral **610**, a pressure relief device **112** is installed into the pocket **116** such that the pressure relief device **112** provides selective fluid communications between the second chamber **110** and an exterior of the body **102** when the pressure relief device **112** is activated at the first predefined pressure.

(32) Turning now to FIG. 7, an exemplary embodiment of a vessel body **702** is shown. Vessel body **702** can include a first metal surface **704** and a second metal surface **706** positioned substantially parallel to each other. In certain embodiments, the first metal surface **704** and the second metal surface **706** can be made of steel. The body **702** can further include a layer of non-metallic material **708** between the first metal surface **704** and the second metal surface **706**. The non-metallic material **708** can include a single layer or multiple layers, and can be constructed of a single material or multiple materials. In an embodiment, the non-metallic material **708** is a portion of a flexible membrane **106** such as a diaphragm, a bag, or a bladder. In certain embodiments, the non-metallic material **708** can melt when the body **702** is exposed to flames or extreme heat. By way of example and not limitation, the non-metallic material **708** can be made of one or more of polypropylene or a rubber, including Butyl, EPDM rubber, or a blend of Butyl and EPDM Rubber. In one embodiment, the non-metallic material **708** includes a first layer made of polypropylene and a second layer made of EPDM rubber. The material of the first metal surface **704**, the second metal surface **706**, and the non-metallic material **708** can be chosen and configured such that when the body **702** is exposed to heat, the first and second metal surfaces **704**, **706** expand due to their coefficient of thermal expansion. The expansion or bending action of the first and second metal surfaces **704**, **706** creates at least one pressure relief route **710** in between at least one of the first metal surface **704** and the non-metallic material **708**, or the second metal surface **706** and the non-metallic material **708**.

(33) The first metal surface **704** can form an interior of a vessel body **702** and the second metal surface **706** can form an exterior of the vessel body **702**. The non-metallic material **708** can be compressed in between the first metal surface **704** and the second metal surface **706**. In certain embodiments, the non-metallic material **708** can be compressed to less than 50% of its uncompressed thickness. In still other embodiments, the non-metallic material **708** can be compressed to less than 20% of its uncompressed thickness. In certain embodiments, the thickness of the non-metallic material can be approximately 0.26 inches when uncompressed, and as thin as 0.05 inches when in a compressed state between the first metal surface **704** and the second metal surface **706**. The first metal surface **704** can be thicker than the second metal surface **706**. During normal operation when there is no fire or excessive heat present, there exists a seal between the

first metal surface **704** and the non-metallic material **708** such that the pressure relief route **710** is non-existent.

(34) In certain embodiments, the first surface **704** can have a curved shape. In this embodiment, the non-metallic material **708** can have a corresponding curved shape to create a seal during normal operation. The non-metallic material **708** can also have a curved shape that is a mirrored image of the curved shape of the first surface **704**. The first metal surface **704** can be designed to be more rigid than the second metal surface **706**. This feature can be accomplished either by designing the first metal surface **704** to have a stronger shape (e.g. curved, arched, concave) than the second metal surface **706**, constructing the first metal surface **704** from a stronger material than the second metal surface **706**, or by constructing the first metal surface to have a greater thickness than the second metal surface **706**. In such a configuration, the second metal surface **706** bends outwards when heated by an external heat source, which causes the curvature of the non-metallic material **708** to relax. The relaxation of the curvature of the non-metallic material **708** loosens the seal between the first metal surface **704** and the non-metallic material **708**, and eventually creates a pressure relief route **710** between the first metal surface **704** and the non-metallic material **708**. The pressure relief route **710** can lead directly to an exterior of the vessel **700** or to a pressure relief device **714**, which can then vent the contents of the vessel **700** to the exterior of the vessel **700**. It should be appreciated that the non-metallic material **708** can be designed to act as a thermal insulator, which further accentuates a thermal expansion difference between first metal surface **704** and second metal surface **706**.

(35) In one embodiment, the non-metallic material **708** (e.g. a flexible membrane, or rigid liner) is crimped between the first metal surface **704** and the second metal surface **706** at a joint **712**. During normal operation of the vessel **700**, the joint **712** creates a seal so that the contents of the vessel **700** cannot escape from the interior of the vessel **700**. When heat is applied to the body **702** that causes the first and/or second metal surface **704**, **706** to expand and/or bend near the joint **712**, the joint **712** can then open to provide an entrance for the contents of the vessel **700** to escape through the at least one pressure relief route **710**. The pressure relief route **710** can lead directly to an exterior of the vessel **700** or to a pressure relief device **714**, which can then vent the contents of the vessel **700** to the exterior of the vessel **700**.

(36) It should be appreciated that the pressure relief route **710** can be created in various ways. For example, if the vessel **700** body **702** is subject to rapid external thermal activity (e.g. 1200 degrees Fahrenheit in 2 minutes or less), the sealing surfaces between the first and second metal surfaces **704**, **706**, and the non-metallic material **708** is relaxed due to differences in thermal expansion between the materials and/or softening of the non-metallic material **708**, resulting in the opening and formation of the pressure relief route **710**. In another example, slower external thermal activity can cause the non-metallic material **708** to melt, and therefore open and form the pressure relief route **710** between the first metal surface **704** and the second metal surface **706**.

(37) In another embodiment, the vessel **700** body **702** can include a meltable component. In one example, the valve coupled with body **702** and utilized for filling and/or dispensing of contents of the vessel **700** can be constructed of a meltable material. In this manner, if the vessel **700** is exposed to excess external heat, the valve or other meltable component will melt and provide a path for the pressurized contents of the vessel **700** to escape.

(38) Moreover, the word “exemplary” is used herein to mean serving as an example, instance or illustration. Any aspect or design described herein as “exemplary” is not necessarily to be construed as advantageous over other aspects or designs. Rather, use of the word exemplary is intended to present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or.” That is, unless specified otherwise, or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. Further, “at least one of A and

B”, or “at least one of A or B” and/or the like generally means A or B or both A and B. In addition, the articles “a” and “an” as used in this application and the appended claims may generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

(39) Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features, ranges, and acts described above are disclosed as example forms of implementing the claims.

(40) Also, although the disclosure has been shown and described with respect to one or more implementations, equivalent alterations and modifications will occur to others skilled in the art based upon a reading and understanding of this specification and the annexed drawings. The disclosure includes all such modifications and alterations and is limited only by the scope of the following claims. In particular regard to the various functions performed by the above described components (e.g., elements, resources, etc.), the terms used to describe such components are intended to correspond, unless otherwise indicated, to any component which performs the specified function of the described component (e.g., that is functionally equivalent), even though not structurally equivalent to the disclosed structure which performs the function in the herein illustrated exemplary implementations of the disclosure. In addition, while a particular feature of the disclosure may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application. Furthermore, to the extent that the terms “includes,” “having,” “has,” “with,” or variants thereof are used in either the detailed description or the claims, such terms are intended to be inclusive in a manner similar to the term “comprising.”

(41) The implementations have been described, hereinabove. It will be apparent to those skilled in the art that the above methods and apparatuses may incorporate changes and modifications without departing from the general scope of this invention. It is intended to include all such modifications and alterations in so far as they come within the scope of the appended claims or the equivalents thereof.

Claims

1. A vessel, comprising: a body that defines an interior space, wherein the body comprises: a first metal surface; a second metal surface; and a non-metallic material crimped between the first metal surface and the second metal surface at a joint; and a pressure relief device coupled to the body, wherein the pressure relief device is configured to vent contents of the vessel to an exterior of the body; wherein at least one of: the second metal surface is configured to expand or bend, upon exposure of the vessel to a temperature for a period of time, to create a pressure relief route between the first metal surface and the second metal surface to provide fluid communication between the interior space and the pressure relief device, or the non-metallic material is configured to melt, contract, or soften, upon exposure of the vessel to the temperature for the period of time, to create the pressure relief route.

2. The vessel of claim 1, wherein upon exposure of the vessel to the temperature for the period of time, the second metal surface is configured to expand or bend such that the joint opens to provide an entrance for the contents of the vessel to escape through the pressure relief route.

3. The vessel of claim 1, wherein the second metal surface forms an exterior surface of the body.

4. The vessel of claim 1, wherein the second metal surface is steel.

5. The vessel of claim 1, wherein the first metal surface is thicker than the second metal surface.

6. The vessel of claim 1, wherein the non-metallic material is compressed to less than 50% of its

uncompressed thickness.

7. The vessel of claim 6, wherein the non-metallic material is compressed to less than 20% of its uncompressed thickness.

8. The vessel of claim 1, wherein the first metal surface is more rigid than the second metal surface.

9. The vessel of claim 1, wherein the non-metallic material includes a flexible membrane that separates the interior space into a first chamber and a second chamber.

10. The vessel of claim 1, wherein the non-metallic material comprises one or more of polypropylene, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, or a blend of butyl and EPDM rubber.

11. The vessel of claim 10, wherein the non-metallic material comprises a first layer made of polypropylene and a second layer made of butyl rubber or EPDM rubber.

12. The vessel of claim 1, wherein the temperate is 1200 degrees Fahrenheit and the period of time is 2 minutes or less.

13. A vessel, comprising: a body that defines an interior space, wherein the body comprises: a first metal surface; a second metal surface; and a non-metallic material crimped between the first metal surface and the second metal surface at a joint; and wherein the second metal surface is configured to expand or bend, upon exposure of the vessel to a temperature for a period of time, such that the joint opens to provide an entrance for a contents of the vessel to flow from the interior space through a pressure relief route between the first metal surface and the second metal surface.

14. The vessel of claim 13, further comprising: a pressure relief device coupled to the body, wherein the pressure relief route provides fluid communication between the interior space and an exterior of the body via the pressure relief device.

15. The vessel of claim 13, wherein the second metal surface forms an exterior surface of the body.

16. The vessel of claim 13, wherein the first metal surface is thicker than the second metal surface.

17. The vessel of claim 13, wherein the first metal surface is more rigid than the second metal surface.

18. The vessel of claim 13, wherein the non-metallic material comprises one or more of polypropylene, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, or a blend of butyl and EPDM rubber.

19. The vessel of claim 1, wherein the temperate is 1200 degrees Fahrenheit and the period of time is 2 minutes or less.

20. A vessel, comprising: a body that defines an interior space, wherein the body comprises: a first metal surface; a second metal surface exterior to the first metal surface; and a non-metallic material crimped between the first metal surface and the second metal surface at a joint, wherein the non-metallic material includes: a flexible membrane that separates the interior space into a first chamber and a second chamber; and a liner; wherein the second metal surface is configured to expand or bend, upon exposure of the vessel to a temperature for a period of time, such that the joint opens to provide an entrance for a contents of the vessel to flow from the interior space through a pressure relief route between the first metal surface and the second metal surface.
